TEAM 25

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Day 1 PNL

107,674

Pattern recognition is key

Strong pattern → full size

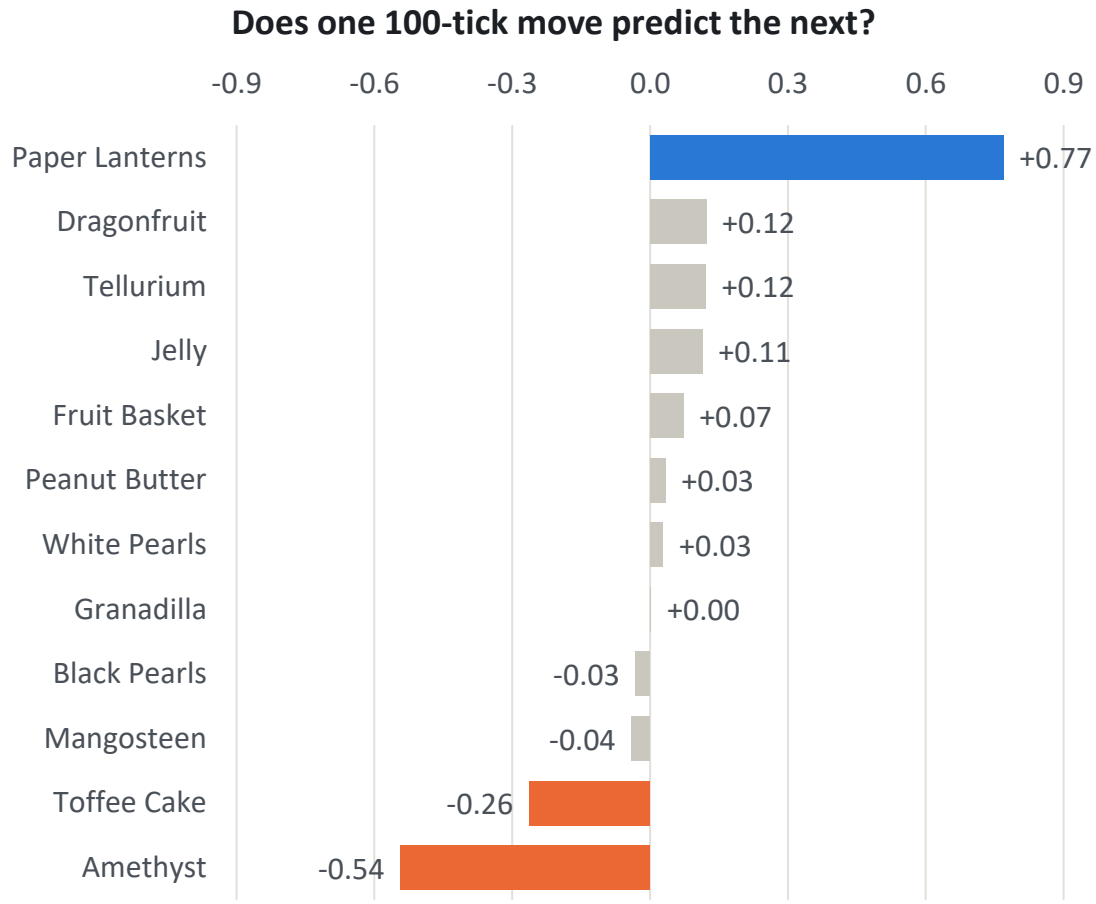
TOFFEE · LANTERNS · 10 vouchers · PINK · BLUE · GOLDEN · AMETHYST

Probable pattern → a sized bet

LYCHEE seems pulled back to ~7,436, so we bet half of what the backtest wanted

No pattern → tiny caps

TELLURIUM · JELLY · PEANUT_BUTTER · fruits · basket · WHITE · BLACK: earn the spread, hold almost nothing



Blue: trends. Orange: snaps back. Grey: no memory. Vouchers and pearls have other patterns (next slides).

Part 1: Core products with one recipe

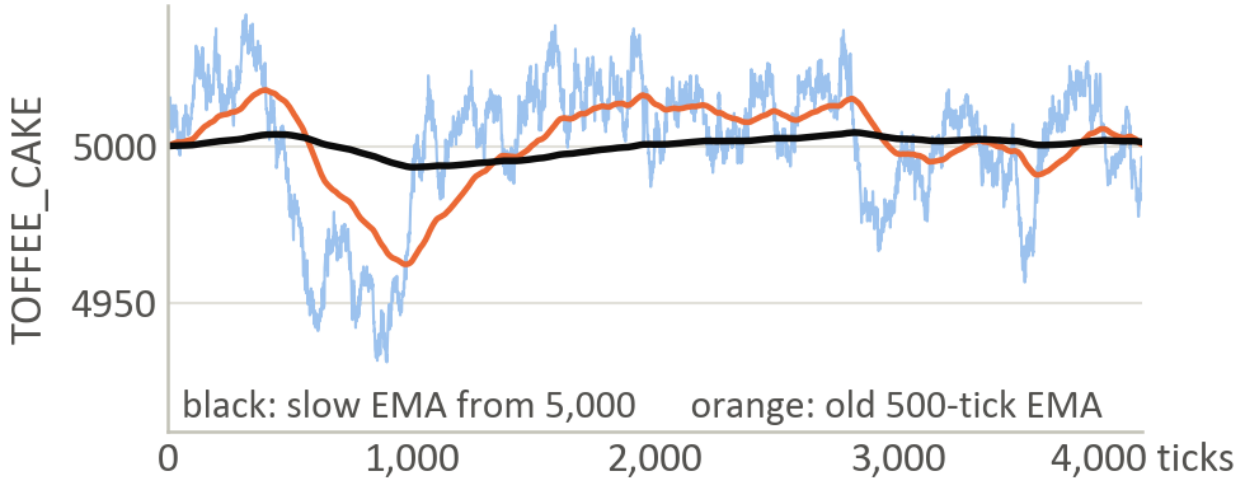
1. **Fair value** — our estimate of the true price
2. **Take** — any quote clearly through it
3. **Make** — join the best bid and ask, or skew away from our position

Product	Pattern	Fair value	Execution	Cap / limit	Profit
TOFFEE_CAKE	snaps back to 5,000	very slow EMA from 5,000	take + skew	**50 / 50**	**65,395**
PAPER_LANTERNS	trends persist	EMA + 0.75 × last 100-tick move	take + join	**50 / 50**	**55,466**
AMETHYST	fixed at 10,000	10,000	take + join	**50 / 50**	**1,613**
TELLURIUM	random, big jumps	none	join only	**6 / 40**	**14,266**
JELLY, PEANUT_BUTTER	random walks	EMA (50 / 20 ticks)	skew only	**3 / 50**	**1,466**
Part I total					**138,206**

Cap: the most we'll hold. A cap far below the exchange limit means there's no edge, so we hold little and just earn the spread.

TOFFEE snaps back and LANTERNS keeps trending

TOFFEE_CAKE · mean reversion

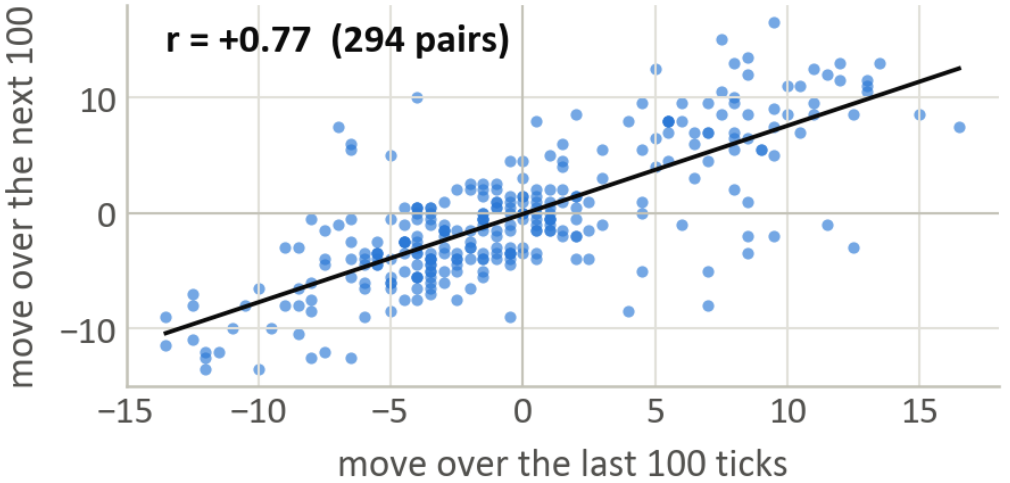


Light blue: price. It keeps returning to ~5,000. Our first 500-tick average (orange) just chased it.

Fair: slow average that starts at 5,000 → 65.4K

Trade-off: hard-coded 5,000 made 67K, but lost up to 7.3K a day when the level moved 25

PAPER_LANTERNS · momentum



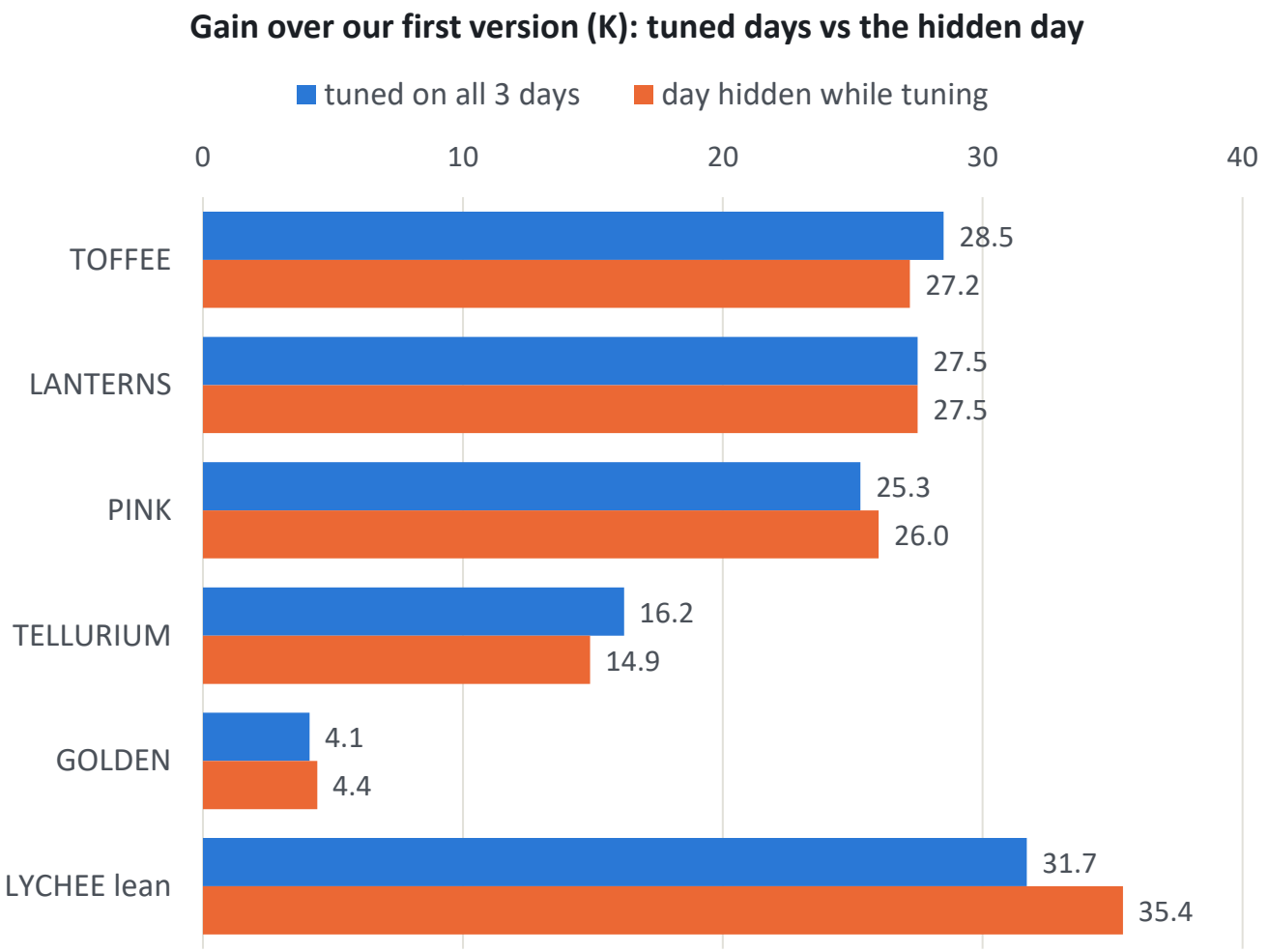
Each dot: two back-to-back 100-tick moves. A rise is usually followed by another rise.

Fair: average + 0.75 × the last 100-tick move → 55.5K

Robust: every weight from 0.5 to 1.0 made 45–55K

Part I total: 138.2K, up from 60.0K in our first version

Challenge 1: with 3 days every tweak looks good



A change ships only if it wins every test

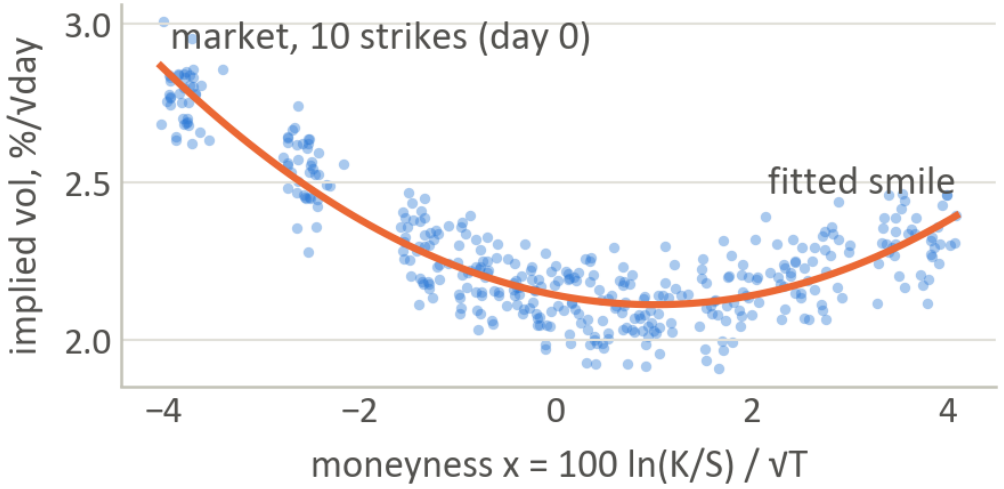
3 days	6 half-days
3-way rotation	nearby settings
stress days	mirrored days
wrong-world days	

Caught: re-tuned voucher settings looked better on the tuning days but lost 11.4K on the hidden days. We kept the old ones.

Rotation: tune on 2 days, test on the hidden 3rd, three times. Orange ≈ blue means the gains are real, not fitted.

Part 2: Ten vouchers use the help of smile

1 · One smile fits all ten strikes



Implied vol vs moneyness, day 0. A strike misses the smile by only ~0.09 vol points.

2 · Price every strike with Black–Scholes

1. **Black–Scholes backwards** on each price → implied vol
   $C = S \cdot \Phi(d_1) - K \cdot \Phi(d_2)$
2. **Fit one smile** through all 10 strikes
   $\sigma(x) = a + b \cdot x + c \cdot x^2, \quad x = 100 \cdot \ln(K/S) / \sqrt{T}$
3. **Fair price** = Black–Scholes at the smile's vol
4. **Trade the gap:** sell 3.5 above fair, buy 3.5 below

197K

from the 10 vouchers: our biggest source

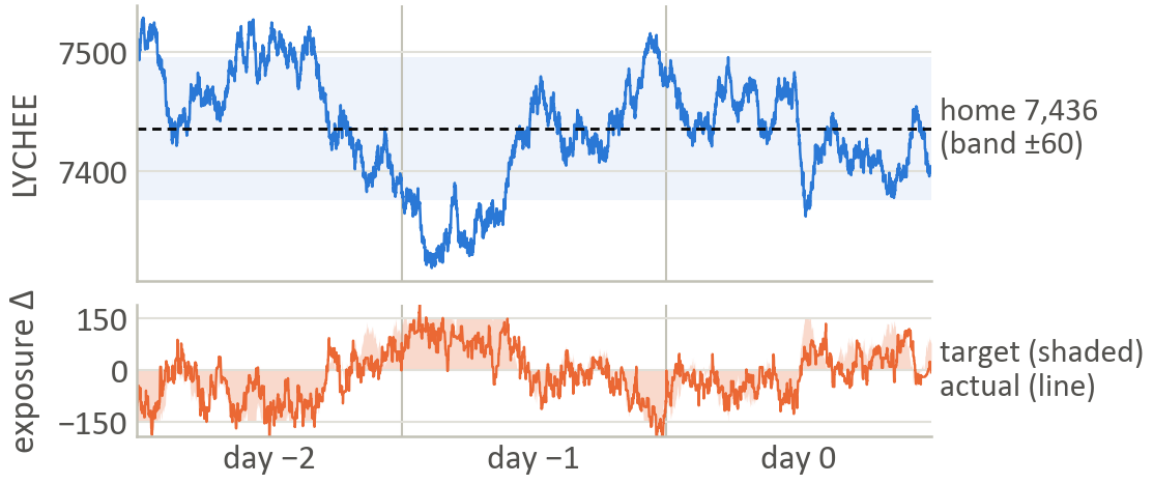
62K vs 49K

a day: one smile vs pricing each strike alone

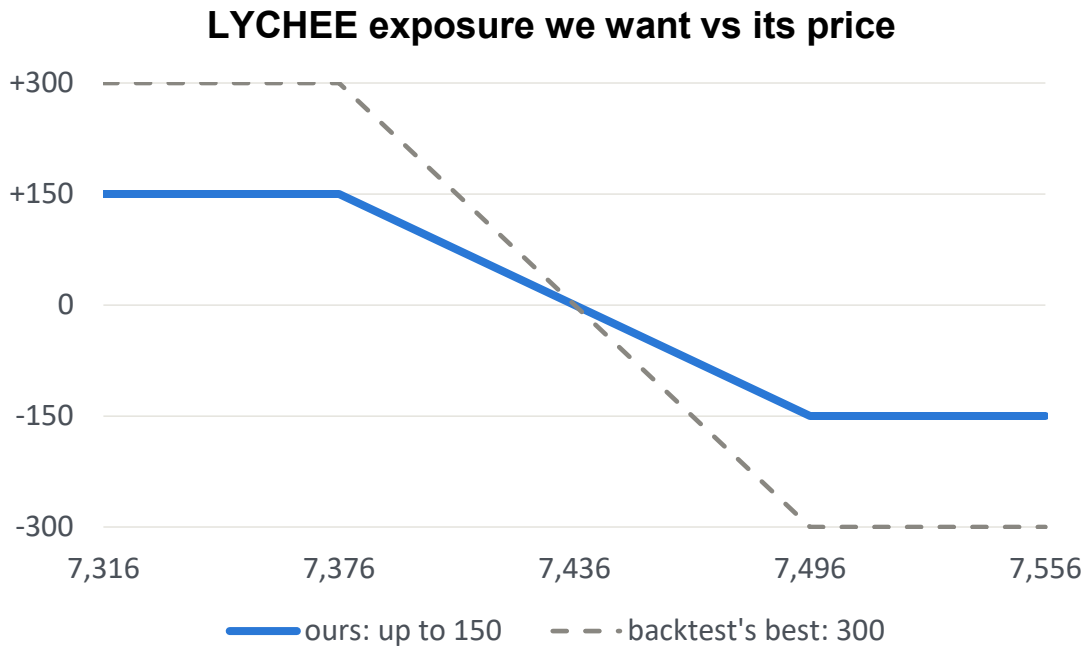
82 → 40

hidden LYCHEE exposure, halved by shading prices (~1K). Hedging: 14–55K

LYCHEE's leash isn't proven so we bet half



Top: LYCHEE kept returning to ~7,436 (band ±60). Bottom: our exposure target (shaded) and actual (line).



150, not 300

300 scored best in the backtest. We halved it

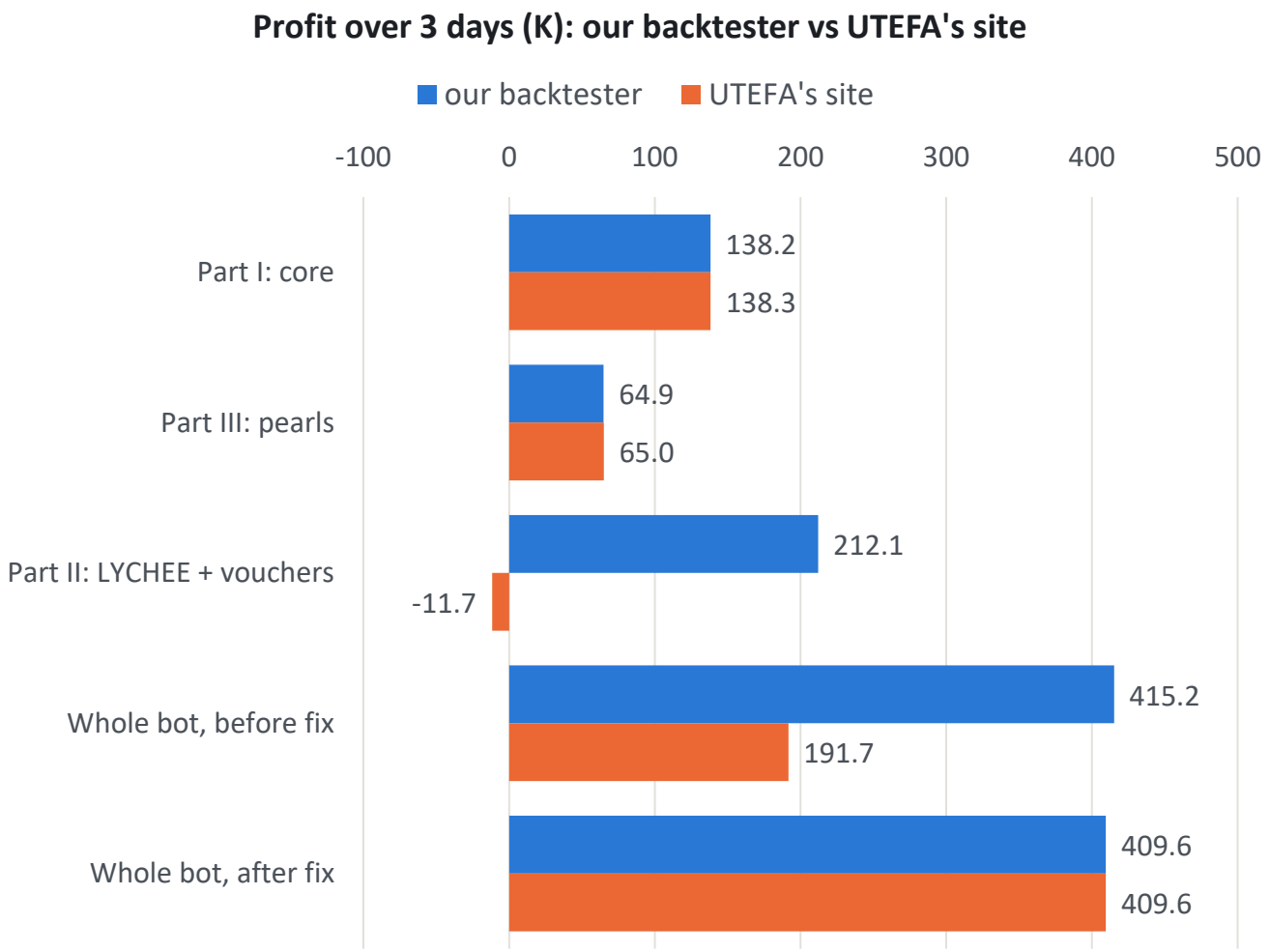
−2.80

ADF test (5% bar: −2.86). 1 in 10 random walks look this leashed

+10–13K / −10–16K

a day if the leash holds / if LYCHEE runs away instead

Challenge 2: the site said 192K, we said 415K



1. Uploaded each part alone: only the LYCHEE part broke (212K here, −12K there)

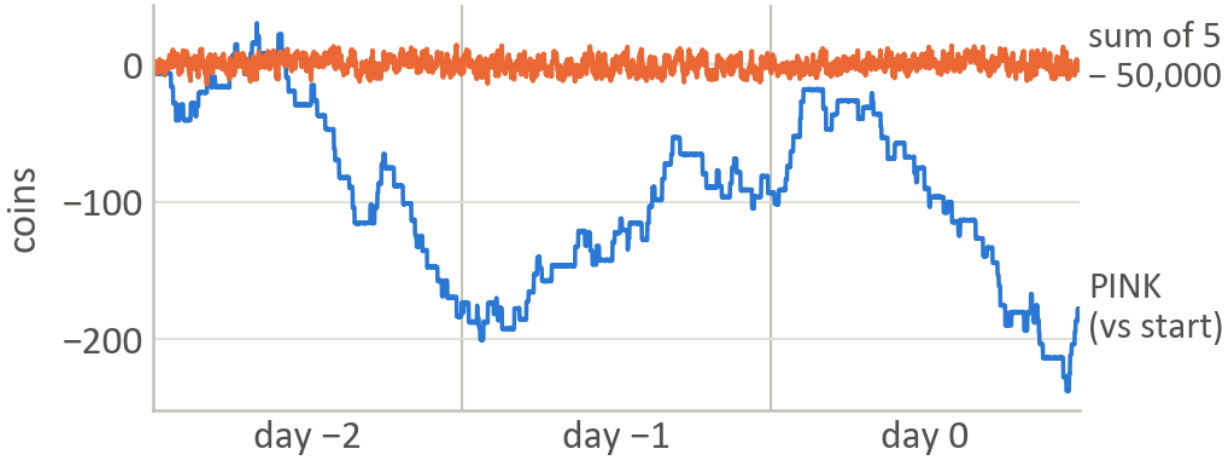
2. Its one unverified input: a trade-flow signal read from other traders' trades

3. Turned it off: 409,590 on both, to the coin. It had been worth just 5.6K

Lesson: never depend on an input you can't verify, and cap every adjustment.

Part 3: PINK closes the 50,000 sum and BLUE saws

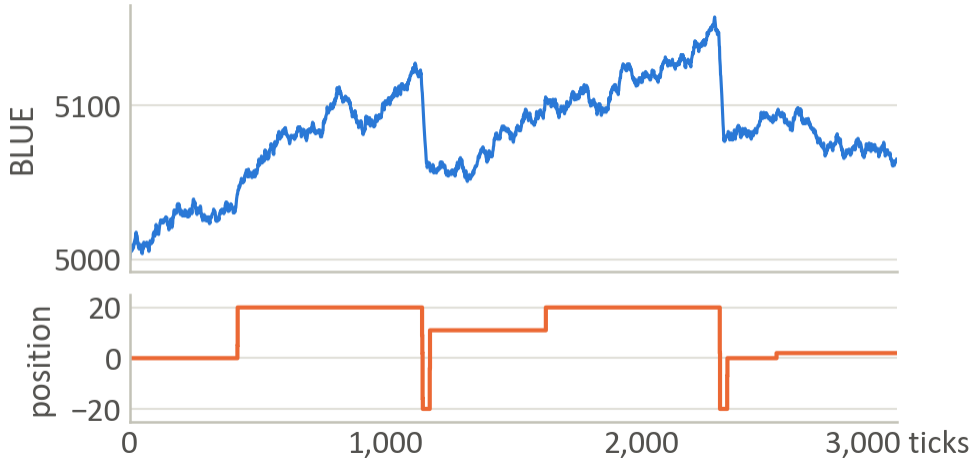
PINK · fair = 50,000 – the other four



PINK wanders 200+ coins (blue). The five pearls' sum − 50,000 stays within a few coins (orange).

27.1K. Only PINK closes the gap. The same idea on the other pearls lost 10–30K each.

BLUE · hold the creep, ride the drop

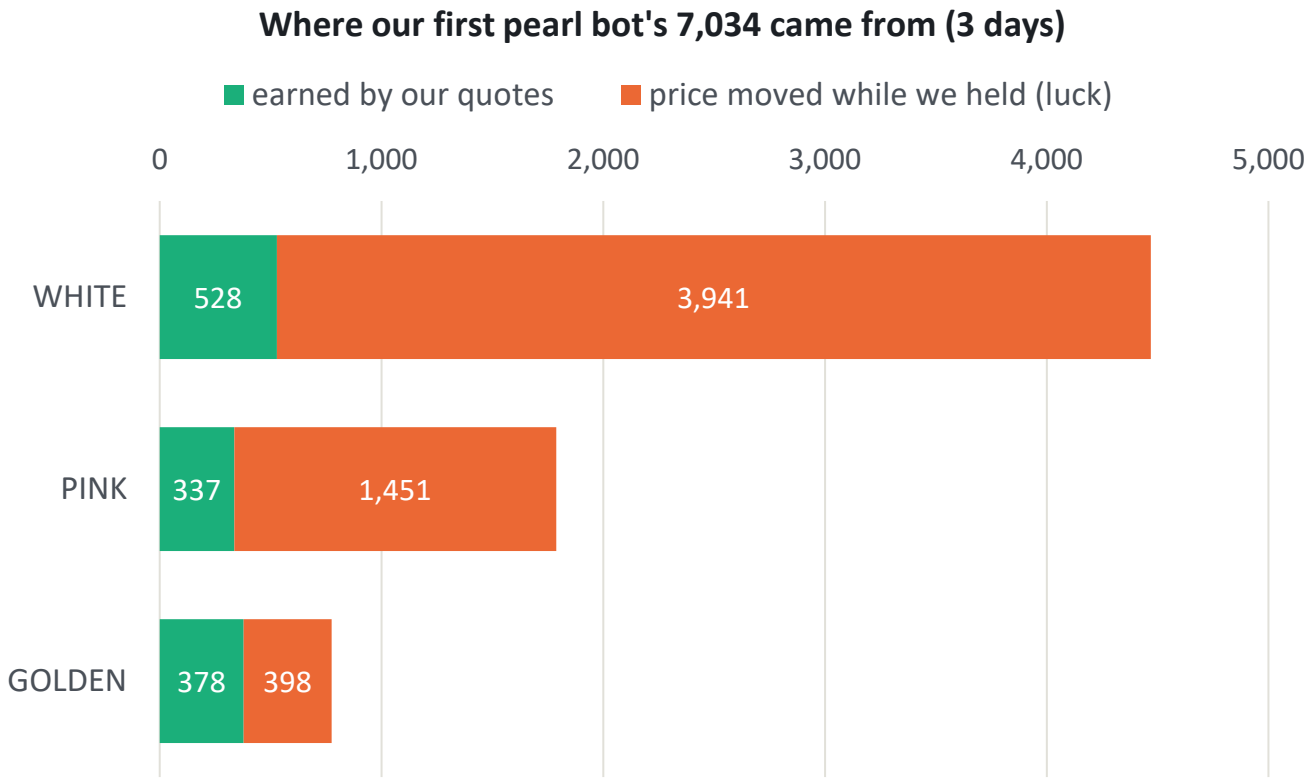


Top: BLUE creeps up, then drops ~45 in under 50 ticks. Bottom: our position.

27.7K. Trade-off: a slower drop detector. 5K less in the backtest, but it loses 2K (not 18–23K) on days with no sawtooth.

Part III total: 64.9K (GOLDEN's step levels 4.9K, WHITE + BLACK 5.1K)

Challenge 3: our first pearl bot's 7K was luck



We split each product's profit into what the quotes earned and what came from holding a position while the price moved.

82% luck

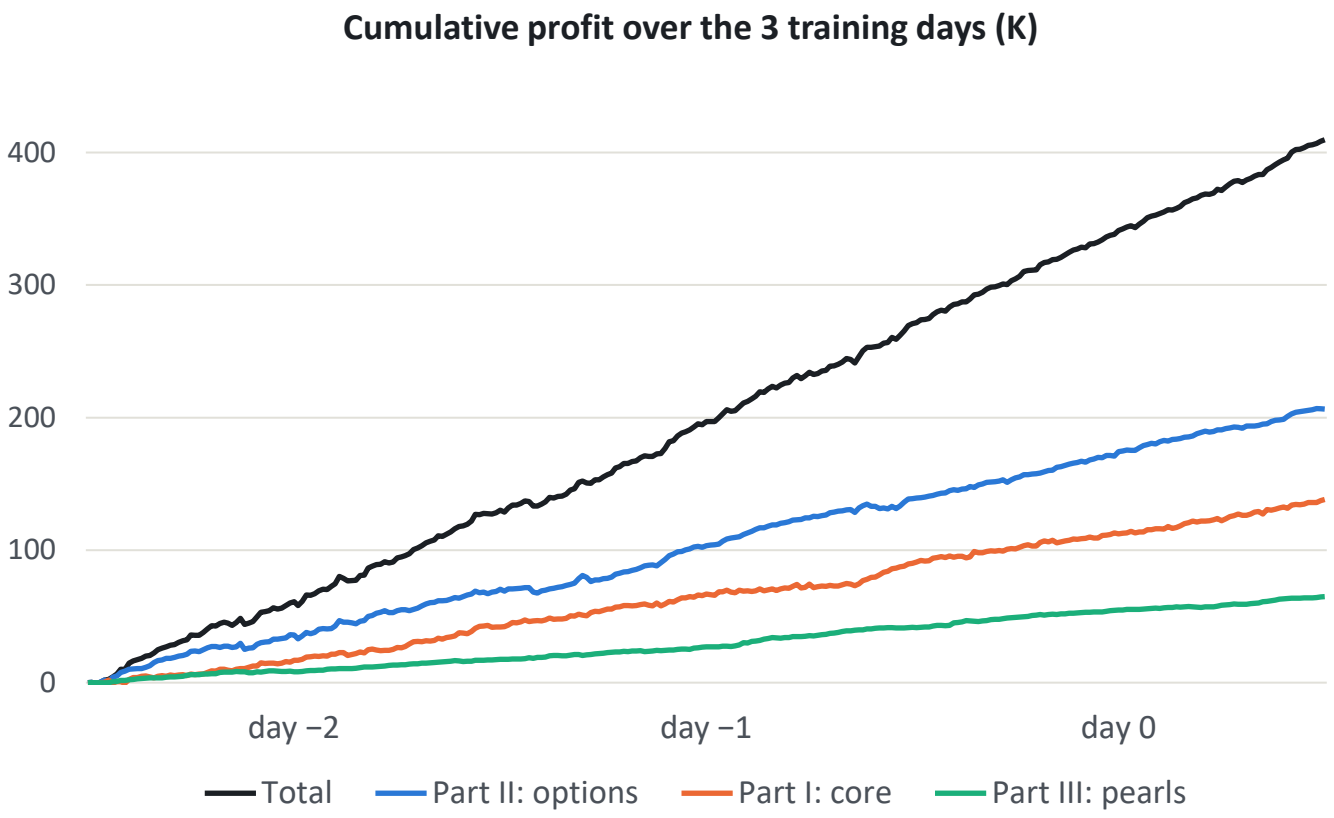
e.g. +3,295 on WHITE in one day, from being short while it fell 460

Fix: split every pearl's profit this way. Build a strategy around any patterns: BLUE's saw, PINK's sum, and GOLDEN's steps. No pattern? Small inventory cap?

7.0K → 64.9K

pearls, positive on all 6 half-days

Steady every day, and we know the worst case



	Day −2	Day −1	Day 0
Profit	130.4K	140.6K	138.5K
Largest drop	6.3K	6.9K	3.4K
Sharpe-style	5.8	6.4	6.6

If we're wrong

- LYCHEE has no leash: −10 to −16K that day
- BLUE's sawtooth vanishes: about −2K
- any other pearl pattern vanishes: at most 2–4K

No position without a measured pattern, and every bet sized to the evidence.

TEAM 25

Thank you.

Arham Aamir · Ali Sher · Sepehr Nassertork · Dev Arun

Backup · profit by product (3 training days)

Product	Day −2	Day −1	Day 0	Total
TOFFEE_CAKE	22,277	21,382	21,736	**65,395**
PAPER_LANTERNS	19,153	18,159	18,154	**55,466**
TELLURIUM	1,274	5,680	7,313	**14,266**
AMETHYST	527	608	478	**1,612**
PEANUT_BUTTER	−346	1,710	−3	**1,362**
JELLY	−488	178	415	**104**
VOUCHER_7400	12,780	15,391	9,910	**38,082**
VOUCHER_7500	11,096	13,399	7,486	**31,981**
VOUCHER_7300	12,353	7,605	7,435	**27,393**
VOUCHER_7600	10,524	8,916	6,738	**26,178**
VOUCHER_7000	5,895	3,361	16,902	**26,158**
VOUCHER_7750	5,716	8,737	4,511	**18,964**
VOUCHER_7200	−1,277	4,024	7,811	**10,558**

Product	Day −2	Day −1	Day 0	Total
VOUCHER_7900	5,213	1,460	1,951	**8,624**
VOUCHER_6800	3,371	1,852	2,366	**7,589**
VOUCHER_8100	780	450	306	**1,536**
LYCHEE	4,021	2,681	2,316	**9,018**
MANGOSTEEN	122	−62	426	**485**
GRANADILLA	30	78	8	**115**
DRAGONFRUIT	44	150	−122	**72**
FRUIT_BASKET	−148	638	−707	**−218**
BLUE_PEARLS	6,133	10,664	10,950	**27,747**
PINK_PEARLS	10,192	8,165	8,741	**27,098**
GOLDEN_PEARLS	125	1,998	2,762	**4,886**
WHITE_PEARLS	773	3,432	265	**4,470**
BLACK_PEARLS	288	−38	400	**650**

Totals: Part I 138,206 · Part II 206,534 · Part III 64,850 · all 409,590 (site-verified)

Backup · trade-offs we made (1 of 2)

Decision	We chose	Why
LYCHEE lean size	150 (not 300, the backtest's favourite)	The leash isn't proven (ADF −2.80). A wrong-way day costs 10–16K at 150; 150 → 200 adds 1.7K on good days but costs 2.5K on bad ones
TOFFEE fair value	Slow average from 5,000 (not hard-coded)	Hard-coded lost up to 7.3K/day when the level shifted 25. The slow average won all 24 test days
BLUE drop detector	10 ticks, trigger 12 (not 5 ticks, trigger 8)	The faster one made +5K in-sample but loses 18–23K on days without the sawtooth, against 2K for ours
Voucher LYCHEE risk	Shade prices (not delta-hedge)	Halves exposure (82 → 40) for ~1K; hedging costs 14–55K. The shade is capped at 5 coins
Basket arbitrage	Rejected; cap 1	The basket-vs-fruits gap drifts by hundreds (t = −1.6). Limits cap any hedge at ~13 baskets
Trade-flow signal	Off (was +5.6K locally)	It reads an input the site fills differently; it broke the voucher book there

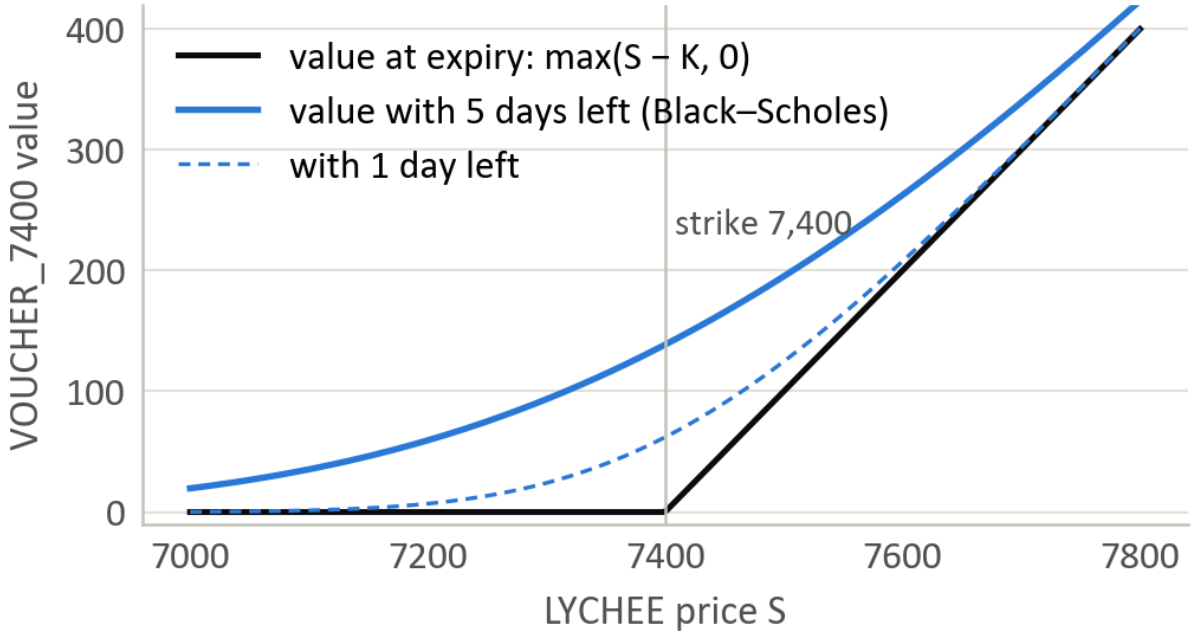
Backup · trade-offs we made (2 of 2)

Decision	We chose	Why
Voucher parameters	Keep the original settings	Re-tuning lost 11.4K on the hidden days: overfitting
PINK sum target	Slow average from 50,000	Hard-coded 50,000 failed 3 of 6 shifted-sum stress days
Random walks	Caps 1–6	No edge means holding is luck. TELLURIUM at cap 40 swung −20K to +25K a day
Voucher fair value	One smile across all strikes	~62K/day vs ~49K/day using each strike's own average vol
How to fade LYCHEE	Through the vouchers	Trading LYCHEE itself: crossing its spread ate the edge (median −7K/day)
Libraries	Plain Python	We couldn't confirm numpy/pandas on the server, and a crash would zero everything

Backup · ideas we tested and dropped

Idea (where it came from)	What we found
Basket vs fruits arbitrage (Prosperity 2/3)	The gap drifts −779 → +900 → −200 and doesn't revert
"Wall-mid" taking (Prosperity 2/3)	Our books have no deep wall. Taking loses 1–4 per unit
Following specific counterparties	Buyer and seller names are always blank
Deep out-of-the-money "bid 0" trick	VOUCHER_8100 trades at 2–44, never 0–1
JELLY / PEANUT_BUTTER pair trade	Correlated (+0.71), but the spread's half-life is 336–2,108 ticks and unstable
Per-strike vol offset on top of the smile	Within noise: 1 in 8 variants would pass by luck
Cubic smile, vega-weighted or leave-one-out fit	No better, or slightly worse
PINK's sum fair value on the other pearls	Loses 10–30K each. Only PINK adjusts
Full 5-pearl arbitrage	Possible only ~20 ticks a day, for ≤ 3 coins
Delta-hedging the vouchers	Costs 14–55K over 3 days
Trade-flow signals (vouchers, TELLURIUM)	TELLURIUM −1.9K. Vouchers +1.4% locally, but broke on the site

Backup · how we price a voucher



VOUCHER_7400 is a call: the right to buy LYCHEE at 7,400. Before expiry it's worth more than max(S − K, 0).

Black–Scholes (r = 0, T in days, σ per √day)

$C = S \cdot \Phi(d_1) - K \cdot \Phi(d_2)$

$d_1 = [\ln(S/K) + \tfrac{1}{2}\sigma^2 T] / (\sigma\sqrt{T}), \quad d_2 = d_1 - \sigma\sqrt{T}$

The smile, refit every tick

$\sigma(x) = a + b \cdot x + c \cdot x^2, \quad x = 100 \cdot \ln(K/S) / \sqrt{T}$

Implied vol by Newton's method (price error ÷ vega)

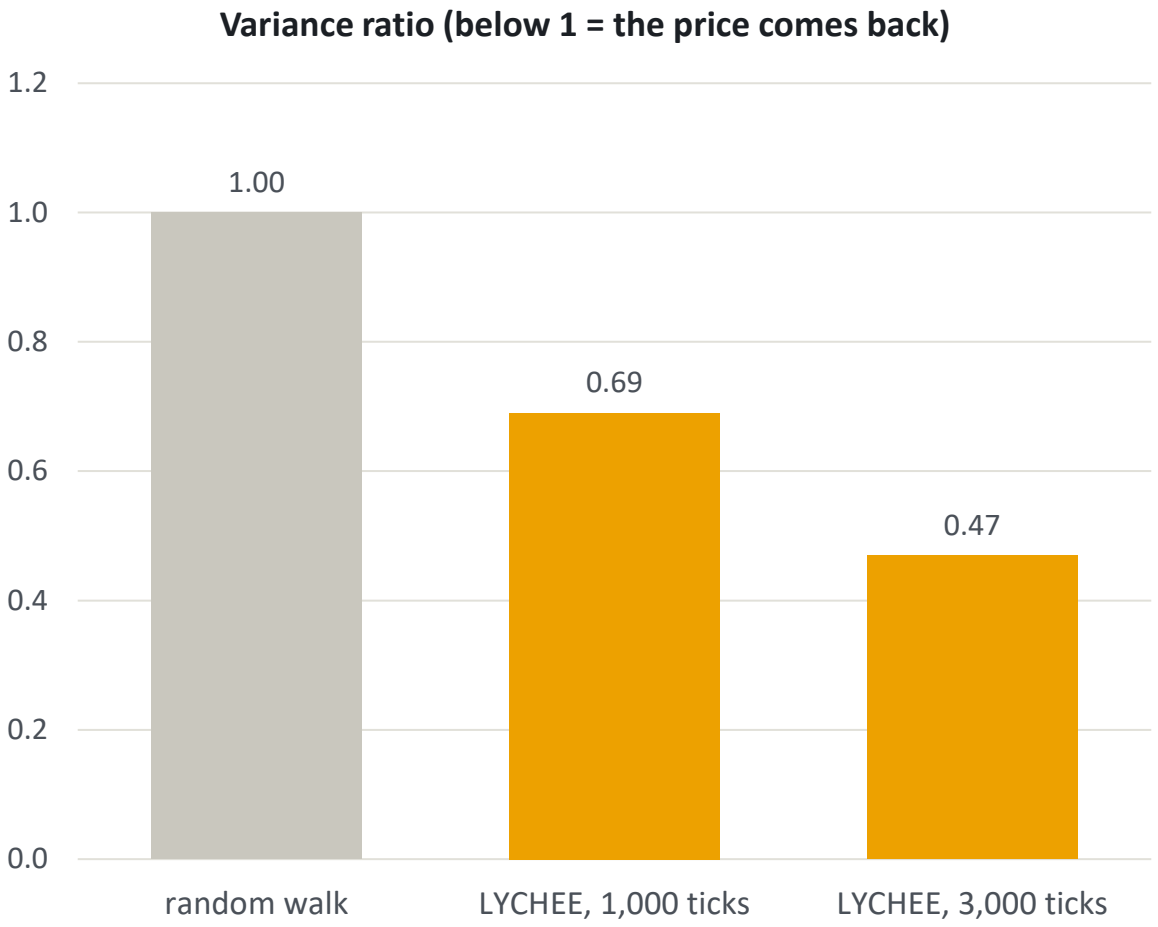
Exposure and the shade

Δ = LYCHEE position + Σ delta × voucher position

fair −= 0.05 × (Δ − target) × delta, capped at 5 coins

target = −150 × clip((S − 7,436) / 60, −1, 1)

Backup · is LYCHEE really leashed?



−2.80

ADF t-stat; the 5% bar is −2.86

1 in 10

random walks look this leashed by luck

7,316–7,530

LYCHEE's whole range over 3 days

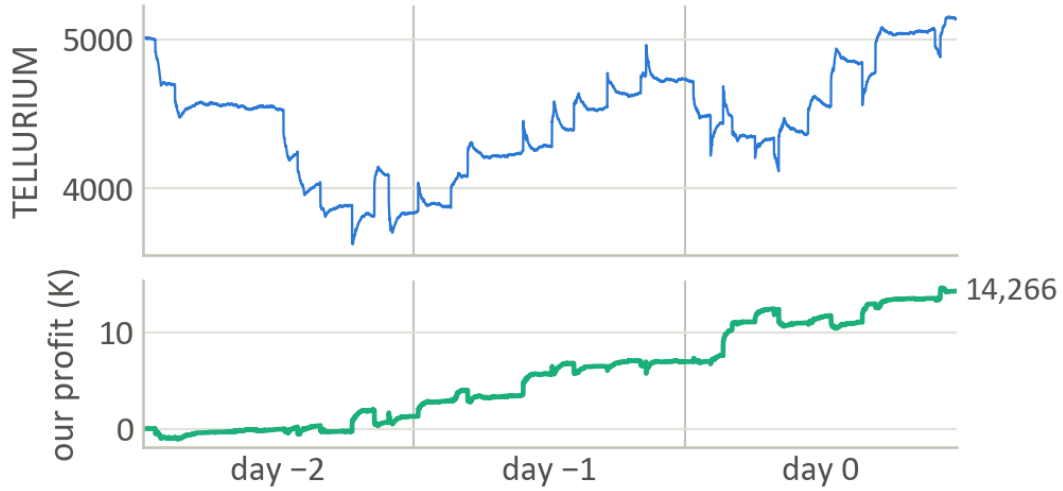
72 / 72

lean settings improved every hidden day

If there's no leash: the lean run backwards lost 10.0K / 12.0K / 15.6K a day; its expected value is about −0.7K a day. If the leash is real: about +11K a day.

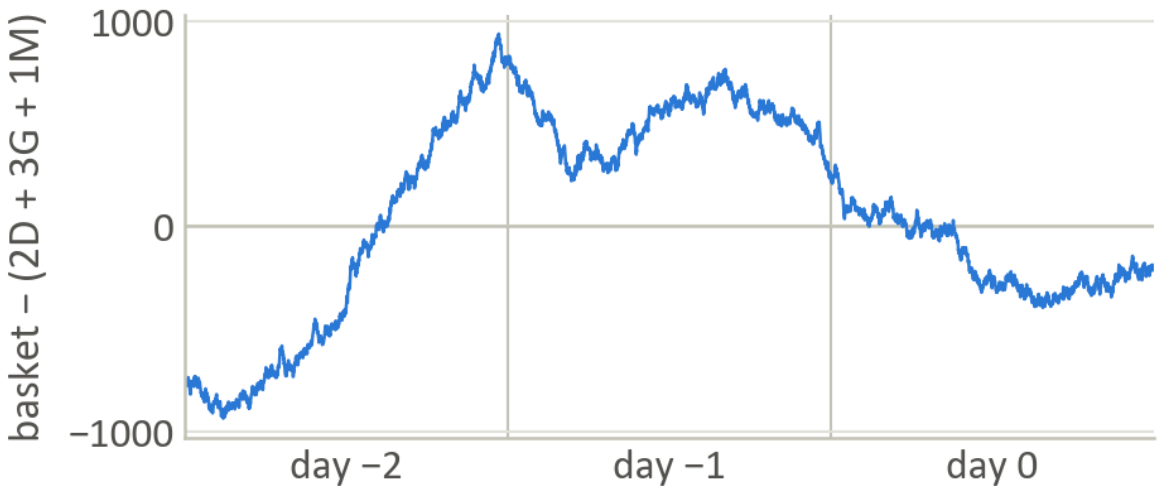
Backup · products with no pattern get tiny caps

TELLURIUM · join only, cap 6 → 14.3K



Top: 100–380-coin jumps. Bottom: our profit with a cap of 6 still climbs. Caps 4–8 were positive every day.

FRUIT_BASKET ≠ 2 DRAGON + 3 GRANADILLA + 1 MANGO

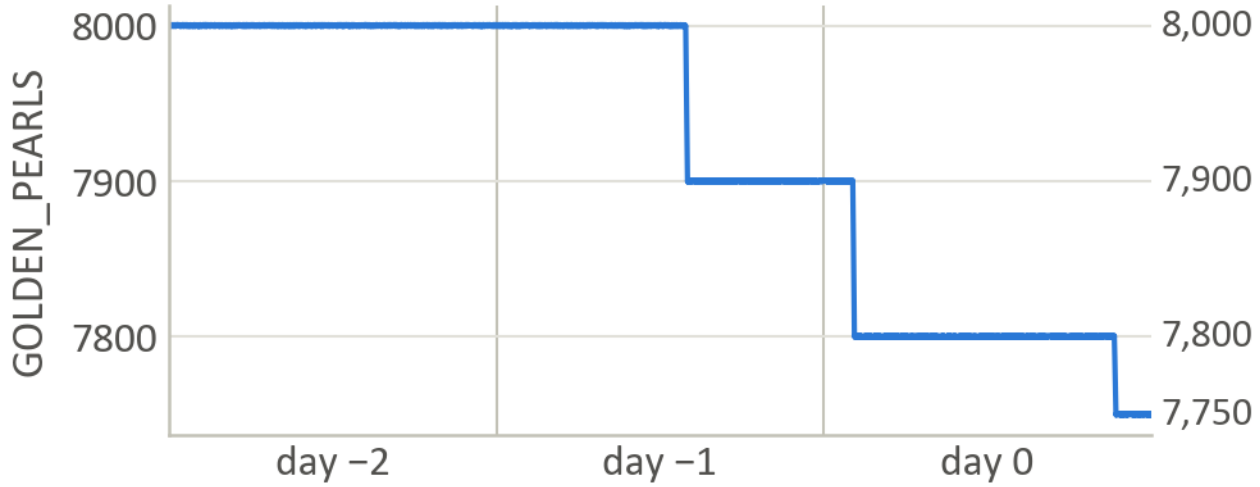


The basket minus its fruits drifts by hundreds instead of snapping back, so there's no safe arbitrage: cap 1.

JELLY and PEANUT_BUTTER: caps cut to 3. Day-to-day swing fell from sd 997 to 360 and from 2,028 to 755, with the same average.

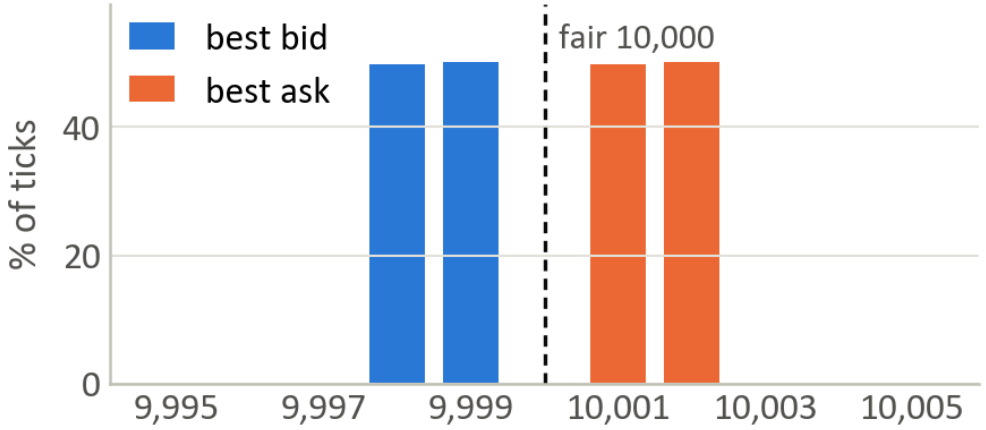
Backup · GOLDEN steps between levels; AMETHYST never moves

GOLDEN_PEARLS · ride the slide → 4.9K



Flat at 8,000 for all of day −2, then slides to 7,900, 7,800, 7,750 at 1–2 coins a tick. Quote around a level; ride a break of 3+.

AMETHYST · fixed at 10,000 → 1.6K



Best bid 9,998–9,999, best ask 10,001–10,002 on every tick. Nothing to take, only a tiny spread to earn.

Backup · every test a change had to pass

1

3 days

better on each training day

2

6 half-days

better on each half of each day

3

3-way rotation

tune on 2 days, test on the hidden 3rd

4

Nearby settings

a plateau, not a lucky peak

5

Stress days

TOFFEE home ±25, level steps, shifted pearl sum

6

Mirrored days

prices flipped upside down

7

Wrong-world days

the pattern removed or reversed

8

Fuzz test

3,000 broken ticks: 0 crashes, 0 bad orders, 0.27 ms a tick